ARTIIS
International Conference on Advanced Research
In Technologies, Information, Innovation
and Sustainability

11-13 & 16 November 2026
Lisbon, Portugal

Artificial Intelligence Methods for Assessing Medical Infrared Thermal Imaging in Early Identification of Thyroid Cancer

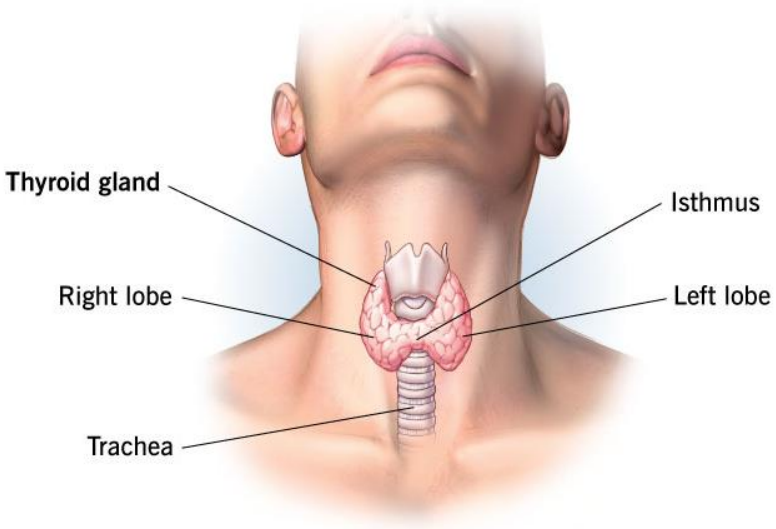
Authors names: Ricardo Vardasca, Rosa M. Cibrian-Ortiz de Anda, Valentina Vassilenko

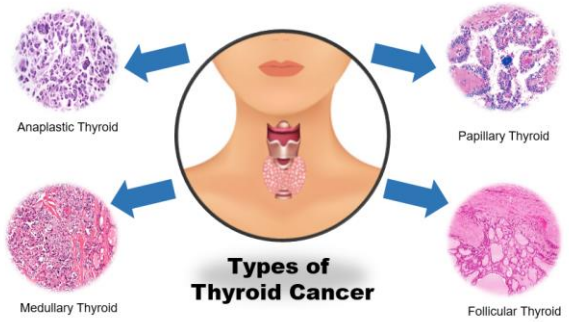
Presenter name: Ricardo Vardasca

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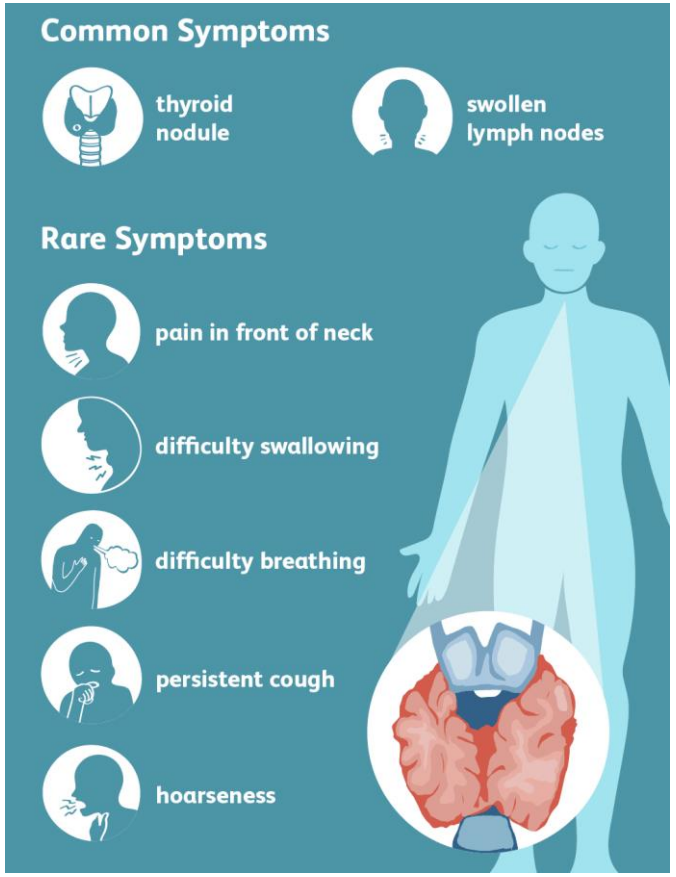
THYROID GLAND

- Produces hormones that regulate metabolism and energy use in almost every cell in the body.
- Cancer facts & figures:
  - 959,281 new cases (ranking 6th globally) and around 48,070 deaths annually, with the highest number of diagnoses occurring across Asia (77.4%) and Europe (8.1%) (GCO, 2026).
  - Three times more common in women than in men (NIH, 2021).
  - The 5-year relative survival rate is 98.3%, reflecting high rates of successful treatment for localized and regional stages (NCI, 2015).
  - Papillary thyroid carcinoma is the most frequent form (accounting for roughly 70% to 84% of cases), followed by follicular, medullary, and anaplastic variants (ENCR, 2024).





THYROID CANCER DIAGNOSIS





Thyroid Ultrasound



Fine-Needle Aspiration



Thyroid Function Tests



Biopsy



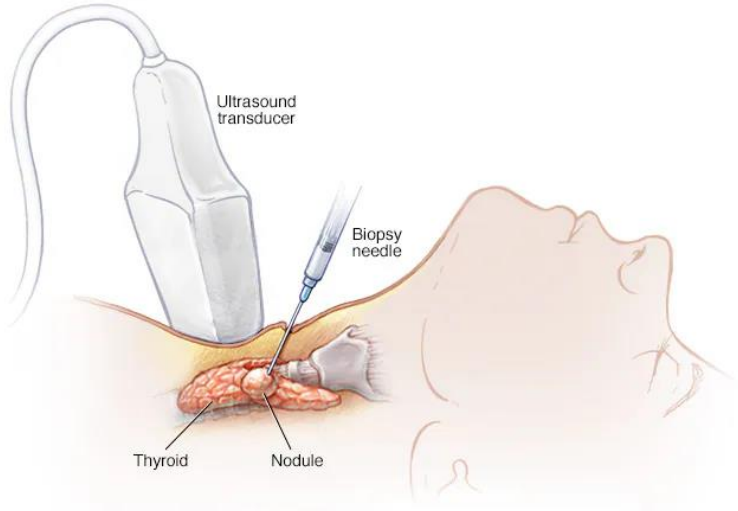
Thyroglobulin Blood Test



Magnetic Resonance Imaging

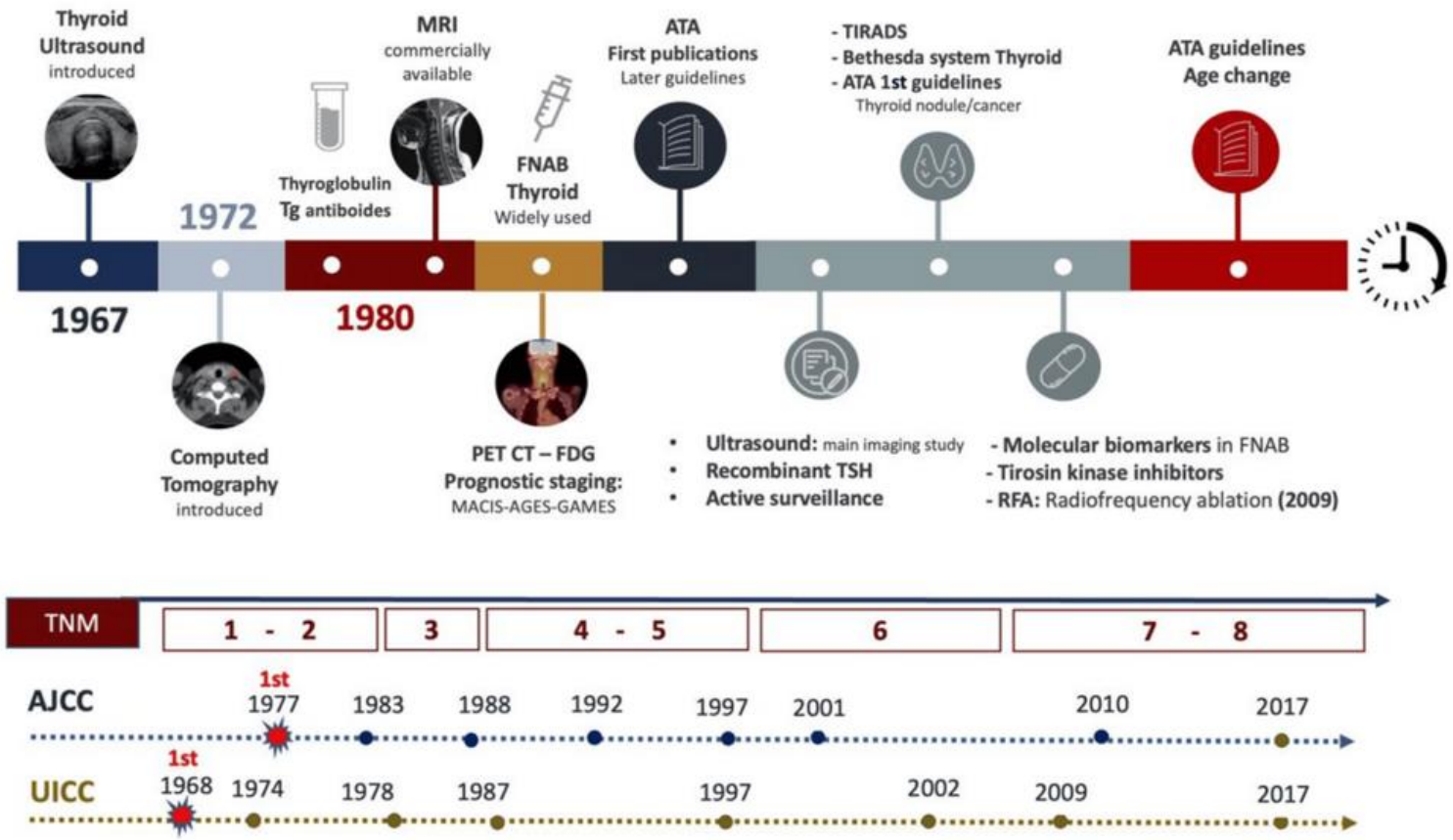


Computed Tomography Scan

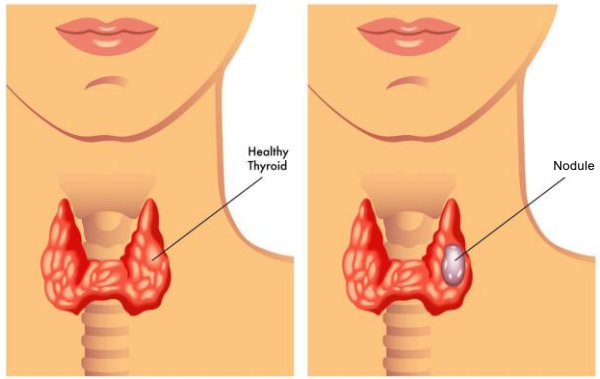


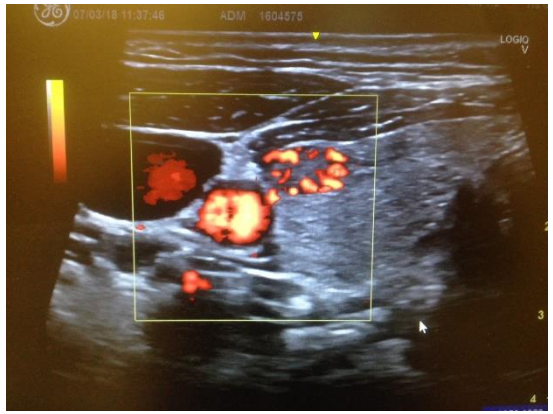
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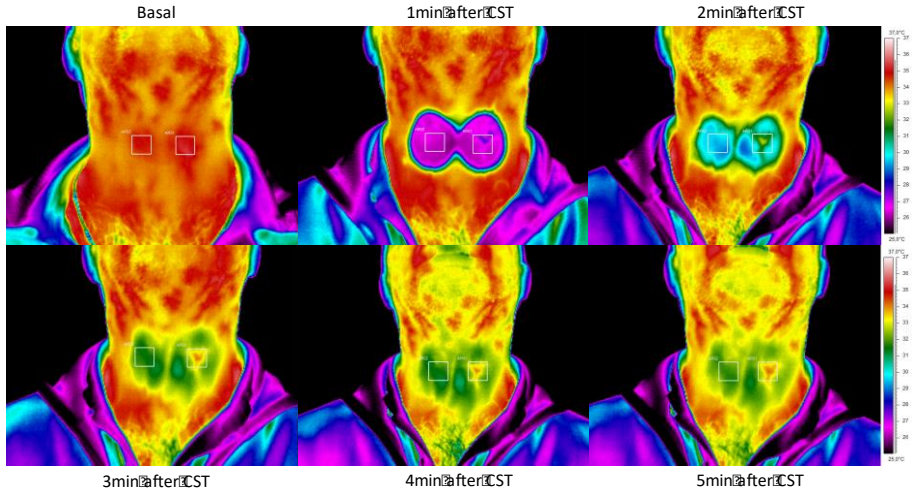
THYROID CANCER DIAGNOSIS



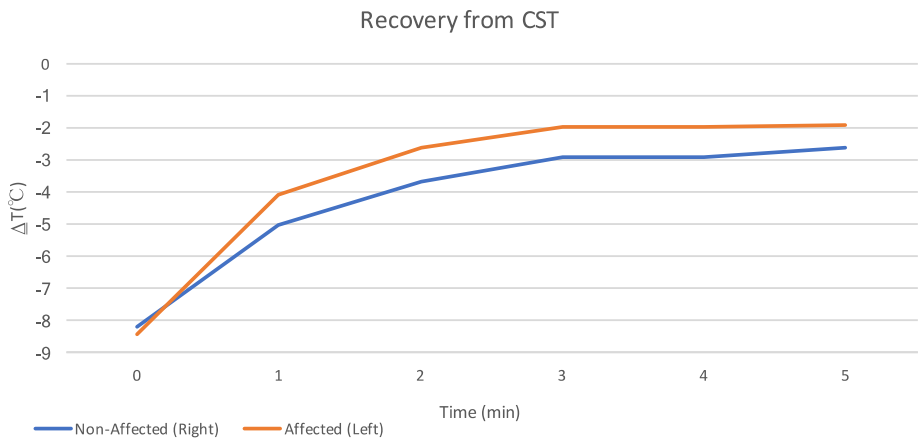
PERSONAL DISCLOSE

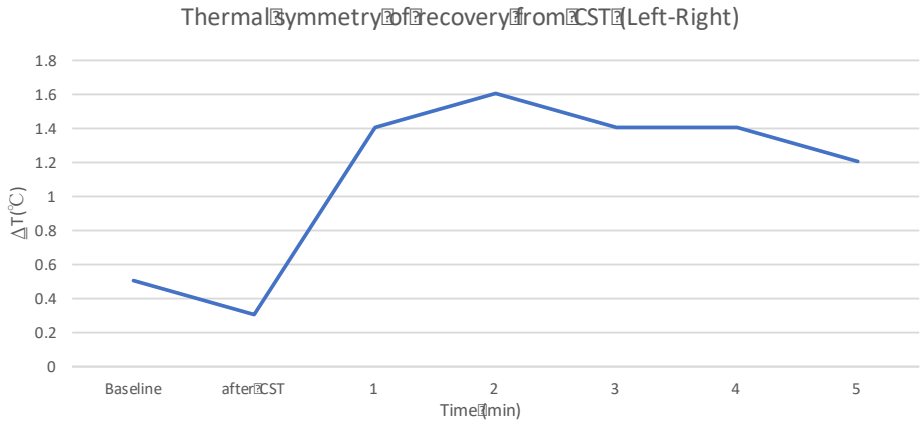












Diagnostic bottlenecks

586,000 Annual Cases

Fastest-rising endocrine malignancy globally with a pronounced 3:1 female predominance (GLOBOCAN 2020).

Aggressive Subtypes

Anaplastic and poorly differentiated carcinomas carry substantial mortality, mandating early intervention.

Screening Latency

Current pathways depend on palpation or incidental discovery, missing early metabolic shifts.

Diagnostic issues

Ultrasound (US) Operator Dependency

Subject to high inter-observer variability and limited functional insights into nodule behavior.

Invasive FNAC (15–30% Indeterminate)

Fine-needle aspiration cytology frequently requires diagnostic hemithyroidectomy for clear diagnoses.

Scintigraphy & PET Limitations

Ionizing radiation exposure and high equipment costs preclude deployment in population-level screening.

The Unmet Diagnostic Need

A non-invasive, radiation-free, objective screening tool capable of pre-selecting high-risk nodules before invasive FNAC procedures are performed.

Biological Mechanisms of Heat Emission

Hypermetabolism & Warburg Effect

Papillary & follicular carcinomas exhibit accelerated aerobic glycolysis, radiating excess metabolic heat.

Pathological Angiogenesis

Disorganized, hyperpermeable tumor microvasculature alters regional skin surface thermal equilibrium.

Benign Thermal Confounders

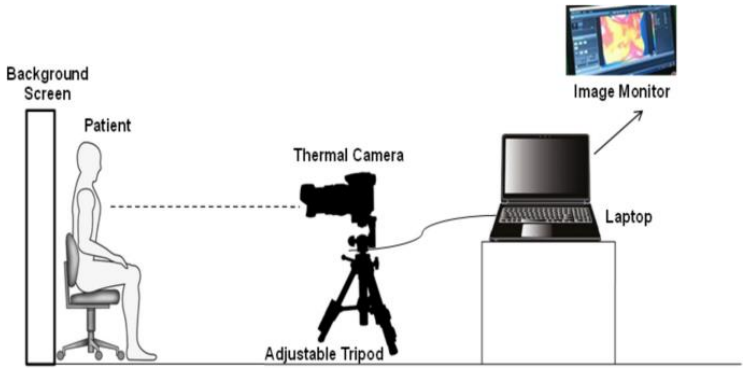
Hashimoto thyroiditis, goiter, and calcifications mimic or obscure malignant thermal signatures.



Physiological Rationale

Thermal alterations often precede structural anatomical changes visible on ultrasound. Automated AI interpretation is essential to detect these subtle physiological signatures against background thermal noise.

Thermographic Acquisition Requirements



LWIR Sensor Calibration

8–14 µm long-wave infrared sensors with thermal sensitivity (NETD) < 50 mK.

Climate Equilibrated Room

Strict ambient control (18–22°C) with a mandatory 10–15 min patient acclimatization.

Multi-Angle Neck Views

Anterior and bilateral oblique captures with standardized neck extension position.

Image Calibration & ROI Extraction

STEP ONE

Sensor Calibration

Non-uniformity correction (NUC), emissivity adjustment, and frame averaging.

Noise Reduction

STEP TWO

Anatomical ROI Segmentation

U-Net & multi-atlas algorithms isolate anterior neck region, excluding carotid artifacts.

Anatomical Delineation

STEP THREE

Bilateral Asymmetry Mapping

Mirror-contralateral subtraction generates temperature difference profiles.

Relative Thermal Index

Feature engineering in a thermal dataset

Texture & Frequency Descriptors

GLCM Statistics: Angular second moment, contrast, correlation, and entropy.

Wavelet Transforms: Multi-resolution frequency band energy distribution maps.

Morphological Properties: Boundary irregularity and thermal hot-spot compactness.

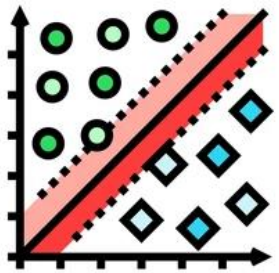
Thermal Symmetry Metrics

Thermal Difference Gradients: Peak-to-peak delta across left/right lobes.

Histogram Skewness & Kurtosis: Quantifying deviation from healthy symmetry.

Isothermal Boundary Index: Surface area ratio of hyperthermal regions.

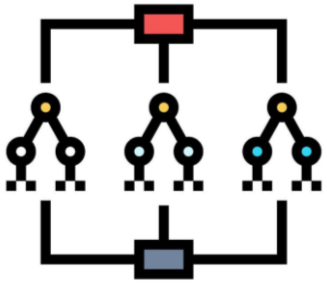
Classical ML classifiers performance benchmarks



Support Vector Machines

RBF kernel SVM outperforms linear variants due to non-linear feature boundary separation.

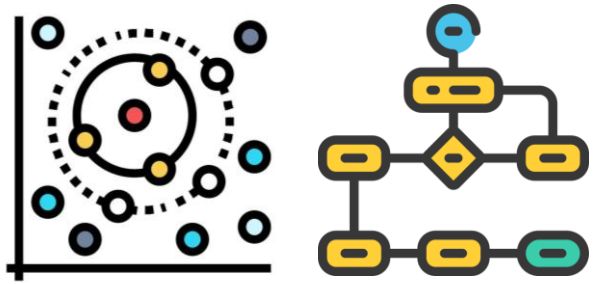
Top ML Classifier



Random Forest Ensembles

Offers strong diagnostic performance alongside intrinsic feature-importance ranking.

High Interpretability



k-NN & Gradient Boosting

Demonstrate lower sensitivity and higher susceptibility to image noise in comparative trials.

Baseline Comparison

Deep Learning: CNNs & Transfer Learning strategies

Pretrained Backbones

VGG-16, ResNet-50, DenseNet-121, and EfficientNet leverage ImageNet feature weights.

Mitigates Data Scarcity

Staged Fine-Tuning

Freezing early layers, updating deeper representations specifically on thermal images.

Best Practice Strategy

Multi-Source Pretraining

Initializing weights on combined thyroid ultrasound datasets reduces domain gaps.

Cross-Modal Transfer

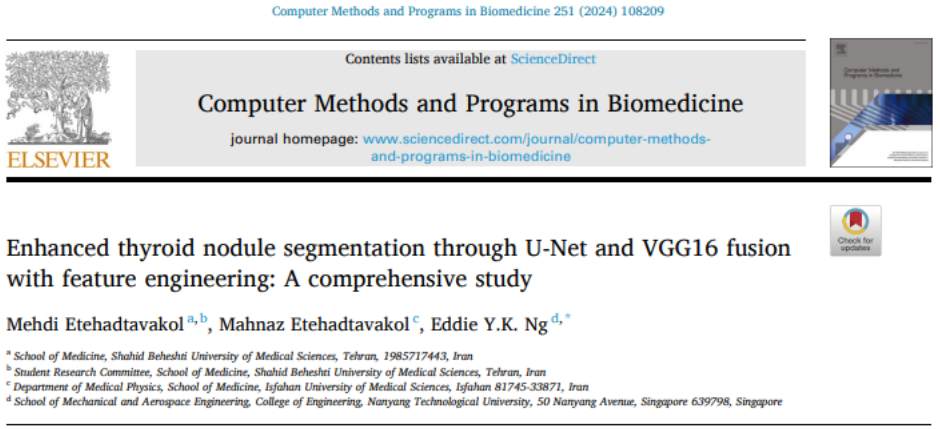
DL validation benchmark

Fused U-Net-VGG16 + Radiomics Landmark (Etehadtavakol 2024)

97.8%

Overall Classification Accuracy

Sub-image entropy feature engineering elevated segmentation Dice scores from 0.08–0.11 to 0.26–0.29 while maintaining ~95% specificity on small patient cohorts (N=46).



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Enhanced thyroid nodule segmentation through U-Net and VGG16 fusion with feature engineering: A comprehensive study

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Interpretability: **Explainable AI Frameworks (XAI)**

Visual Saliency

Grad-CAM and LRP backpropagate activation maps to confirm AI focuses on true thermal anomalies rather than background noise.

Spatial Verification

Feature Attribution

SHAP values quantify Game-Theory contributions of temperature asymmetry and GLCM stats for case reporting.

Quantitative Attribution

Uncertainty Flags

Monte Carlo Dropout estimates prediction variance, automatically routing ambiguous thermograms to human specialists.

Human-in-the-Loop Safeguard

Critical Gaps & Dataset Bottlenecks

Domain Barrier	Current Research Reality	Required Infrastructure Remedy
Dataset Availability	Proprietary cohorts < 300 patients (unlike DMR-IR for breast).	**Public repository (TCIA analogue).**
Methodological Rigor	Optimistic bias from cross-validation without external test sets.	**TRIPOD guideline adherence.**
Class Imbalance	Benign nodules heavily skew training samples away from cancer.	**SMOTE, focal loss & GAN augmentation.**

Point-of-Care & Remote Screening Pathways

Portable Infrared Accessories

Compact thermal sensors coupled with smartphone interfaces for low-cost screening.

Cloud Inference Architecture

Real-time automated image processing returning probabilistic risk scores in seconds.

Demographic Accessibility

Bridges specialist shortages in LMIC environments and rural primary care clinics.



Triage Paradigm

Remote thermographic pre-screening enables priority referral to centralized ultrasound hubs, dramatically improving resource efficiency in overburdened health systems.

Governance: International Regulatory Frameworks



US FDA SaMD Pathway

Software as a Medical Device: Governed under FDA AI/ML Action Plan.

Predetermined Change Control: Protocol for algorithm updates.

Clearance Class: 510(k) or De Novo depending on predicate devices.



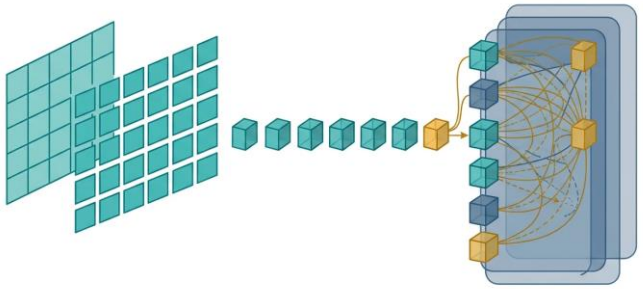
European Union Dual Burden

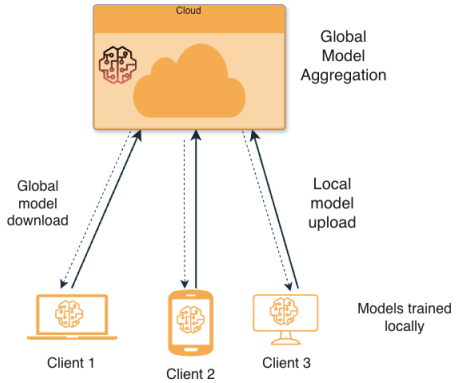
EU MDR 2017/745: Class IIa/IIb requiring prospective PMCF studies.

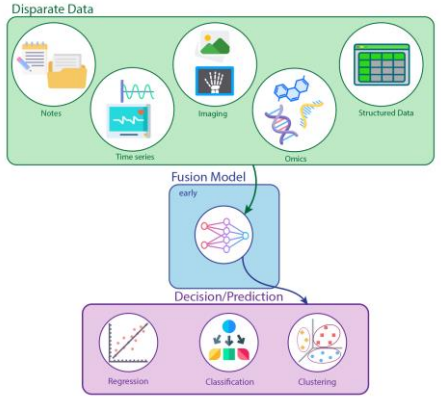
EU AI Act (2024): Annex III "High-Risk" classification enforcement.

Requirements: Cybersecurity, human oversight, and data governance.

Future: Next-Generation Technical Directions







Vision Transformers

Hybrid CNN-ViT models capture global long-range bilateral thermal asymmetries.

Federated Learning

Multi-center model aggregation without centralizing sensitive patient thermal records.

Multimodal Fusion

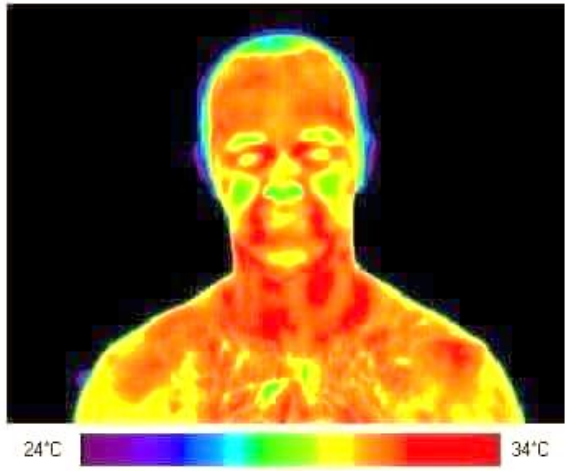
Integrating thermal function + ultrasound structure + liquid biopsy markers.

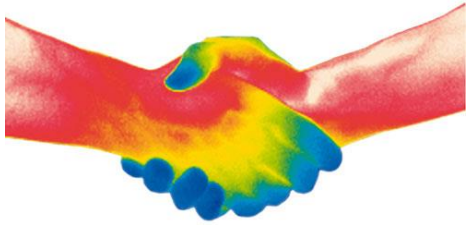
Summary

The Path Toward Standardized, AI-Augmented Thyroid Thermography

AI-enhanced thermography offers a promising non-invasive, contact-free diagnostic window. However, large multi-center prospective validation and public benchmarking datasets are required before routine clinical translation.

Thank you 4 your attention!





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